

Cybersecurity Internship Report

Task 5: SIEM Dashboard Development for Security Event Monitoring Using ELK Stack

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1. Objective

The objective of this task is to design and implement a custom Security Information and Event Management (SIEM) dashboard using the Elastic Stack (ELK) to monitor and visualize critical security events. The dashboard is developed to provide centralized visibility into security incidents such as brute-force attacks, privilege escalation attempts, and data exfiltration activities. Through the use of Elasticsearch and Kibana, security event data is collected, indexed, and transformed into meaningful visualizations that assist security analysts in identifying suspicious behavior, monitoring attack trends, analyzing event severity levels, and tracking source IP activity. The dashboard aims to improve situational awareness, accelerate threat detection, and support efficient security monitoring within a Security Operations Center (SOC) environment.

2. Introduction

Security Operations Centers (SOCs) rely on continuous monitoring and analysis of security events to identify potential threats and respond to incidents effectively. Modern environments generate large volumes of logs and security alerts, making it difficult to detect malicious activities without centralized monitoring and visualization capabilities. Security Information and Event Management (SIEM) platforms address this challenge by aggregating, correlating, and presenting security data in a structured manner. The Elastic Stack (ELK), consisting of Elasticsearch and Kibana, provides a powerful framework for storing, searching, analyzing, and visualizing security events in real time.

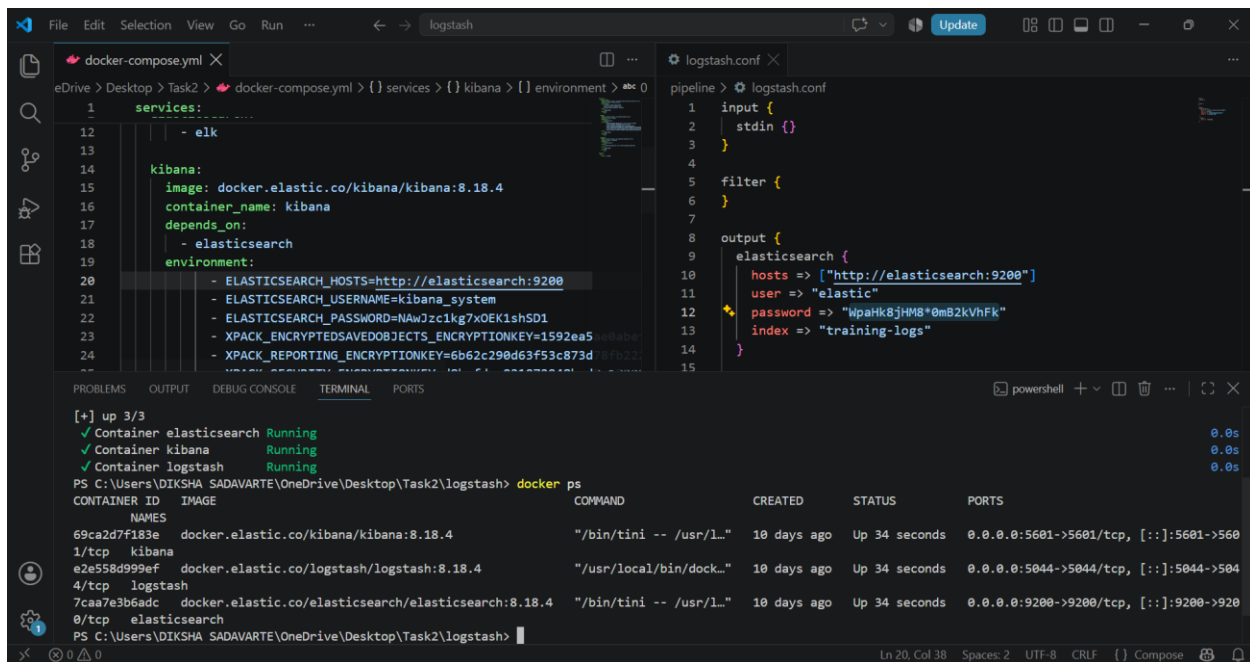
In this project, a custom SIEM dashboard was developed using Kibana to visualize security events associated with brute-force attacks, privilege escalation attempts, and data exfiltration activities. Multiple visualizations, including metric cards, attack distribution charts, severity analysis graphs, source IP monitoring panels, and event tables, were created to provide comprehensive visibility into the security posture of the monitored environment. The dashboard enables analysts to quickly assess ongoing activities, prioritize critical alerts, and support effective incident investigation and response.

3. Tools Used

Tool	Purpose
Elasticsearch	Store and index security event data
Kibana	Create dashboards and security visualizations
Elastic Stack (ELK)	Centralized log management and security monitoring
Windows Operating System	Dashboard development and testing environment
Security Event Dataset	Simulated security events for analysis and visualization

4. Environment Setup

Before designing the SIEM dashboard, the Elastic Stack environment was prepared to provide a centralized platform for storing, indexing, and visualizing security events. Elasticsearch was used as the primary data storage engine, while Kibana was used to create visualizations and dashboards for monitoring security-related activities. A custom index named security-events was created to store simulated security logs representing brute-force attacks, privilege escalation activities, and data exfiltration attempts. After successful data ingestion, a Kibana Data View was configured using the @timestamp field to enable time-based analysis and dashboard visualization. The environment was then validated to ensure that security event data could be successfully retrieved and displayed within Kibana for further monitoring and analysis



The image shows a VS Code editor with two files open: `docker-compose.yml` and `logstash.conf`. The `docker-compose.yml` file defines three services: `elk`, `kibana`, and `logstash`. The `kibana` service is configured to connect to `elasticsearch` at `http://elasticsearch:9200`. The `logstash` service is configured to ingest data from `stdin` and output it to `elasticsearch` at `http://elasticsearch:9200`. The terminal window shows the output of the `docker ps` command, indicating that all three containers are running successfully.

```
services:
  elk:
    image: docker.elastic.co/kibana/kibana:8.18.4
    container_name: kibana
    depends_on:
      - elasticsearch
    environment:
      - ELASTICSEARCH_HOSTS=http://elasticsearch:9200
      - ELASTICSEARCH_USERNAME=kibana_system
      - ELASTICSEARCH_PASSWORD=NawJzc1kg7x0EK1shSD1
      - XPACK_ENCRYPTEDSAVEDOBJECTS_ENCRYPTIONKEY=1592ea5
      - XPACK_REPORTING_ENCRYPTIONKEY=6b62c290d63f53c873d

logstash.conf
pipeline {
  input {
    stdin {}
  }
  filter {
  }
  output {
    elasticsearch {
      hosts => ["http://elasticsearch:9200"]
      user => "elastic"
      password => "WpaHk8jHMB*0mB2kVhFk"
      index => "training-logs"
    }
  }
}
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
69ca2d7f183e	docker.elastic.co/kibana/kibana:8.18.4	"/bin/tini -- /usr/l..."	10 days ago	Up 34 seconds	0.0.0.0:5601->5601/tcp, [::]:5601->5601/tcp
1/tcp kibana	docker.elastic.co/kibana/kibana:8.18.4	"/usr/local/bin/dock..."	10 days ago	Up 34 seconds	0.0.0.0:5044->5044/tcp, [::]:5044->5044/tcp
4/tcp logstash	docker.elastic.co/logstash/logstash:8.18.4	"/bin/tini -- /usr/l..."	10 days ago	Up 34 seconds	0.0.0.0:9200->9200/tcp, [::]:9200->9200/tcp
7caa7e3b6adc	docker.elastic.co/elasticsearch/elasticsearch:8.18.4	"/bin/tini -- /usr/l..."	10 days ago	Up 34 seconds	0.0.0.0:9200->9200/tcp, [::]:9200->9200/tcp
0/tcp elasticsearch	docker.elastic.co/elasticsearch/elasticsearch:8.18.4	"/bin/tini -- /usr/l..."	10 days ago	Up 34 seconds	0.0.0.0:9200->9200/tcp, [::]:9200->9200/tcp

Figure 4.1: Verification of Elasticsearch, Kibana, and Logstash containers running successfully using Docker.

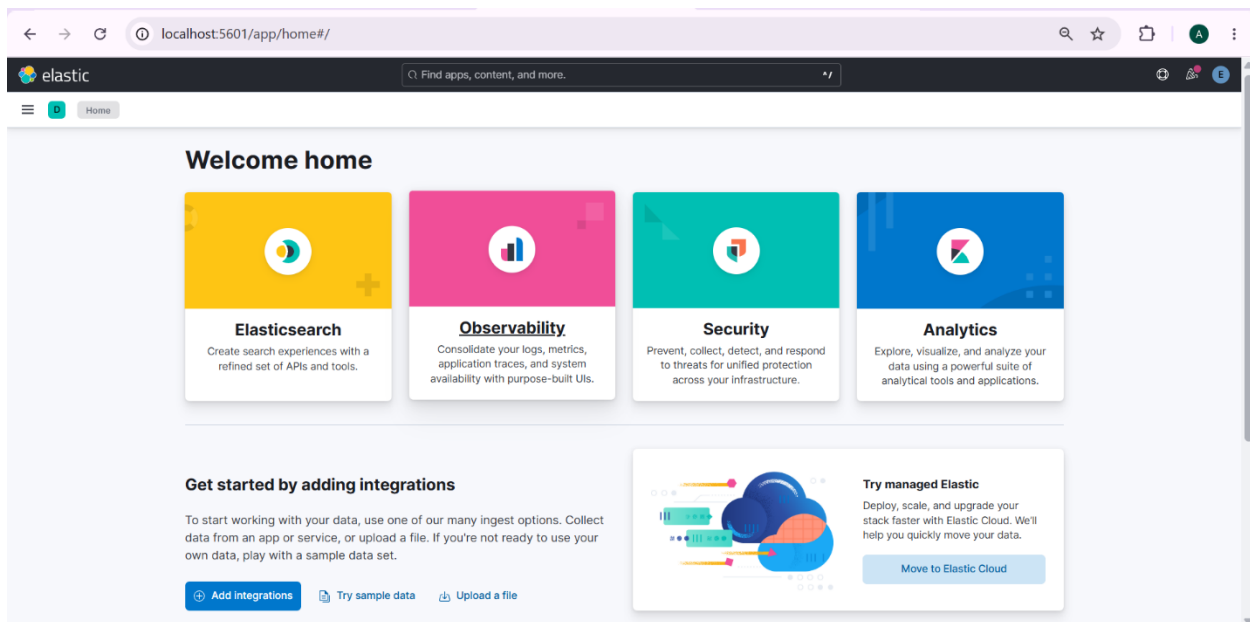


Figure 4.2: Kibana web interface showing successful access to the Elastic Security environment.

The ELK Stack environment was successfully deployed and verified using Docker. Elasticsearch, Kibana, and Logstash containers were confirmed to be running properly, ensuring operational readiness of the platform. Access to the Kibana web interface further validated successful integration of all components and provided a centralized environment for security monitoring, event analysis, detection rule management, and dashboard visualization.

5. Brute Force Attack

Brute-force attacks occur when an attacker repeatedly attempts authentication using multiple password combinations until valid credentials are discovered. Such attacks can lead to unauthorized access, privilege abuse, and compromise of sensitive organizational resources. To detect this activity, a dedicated detection rule was configured within Elastic Security.

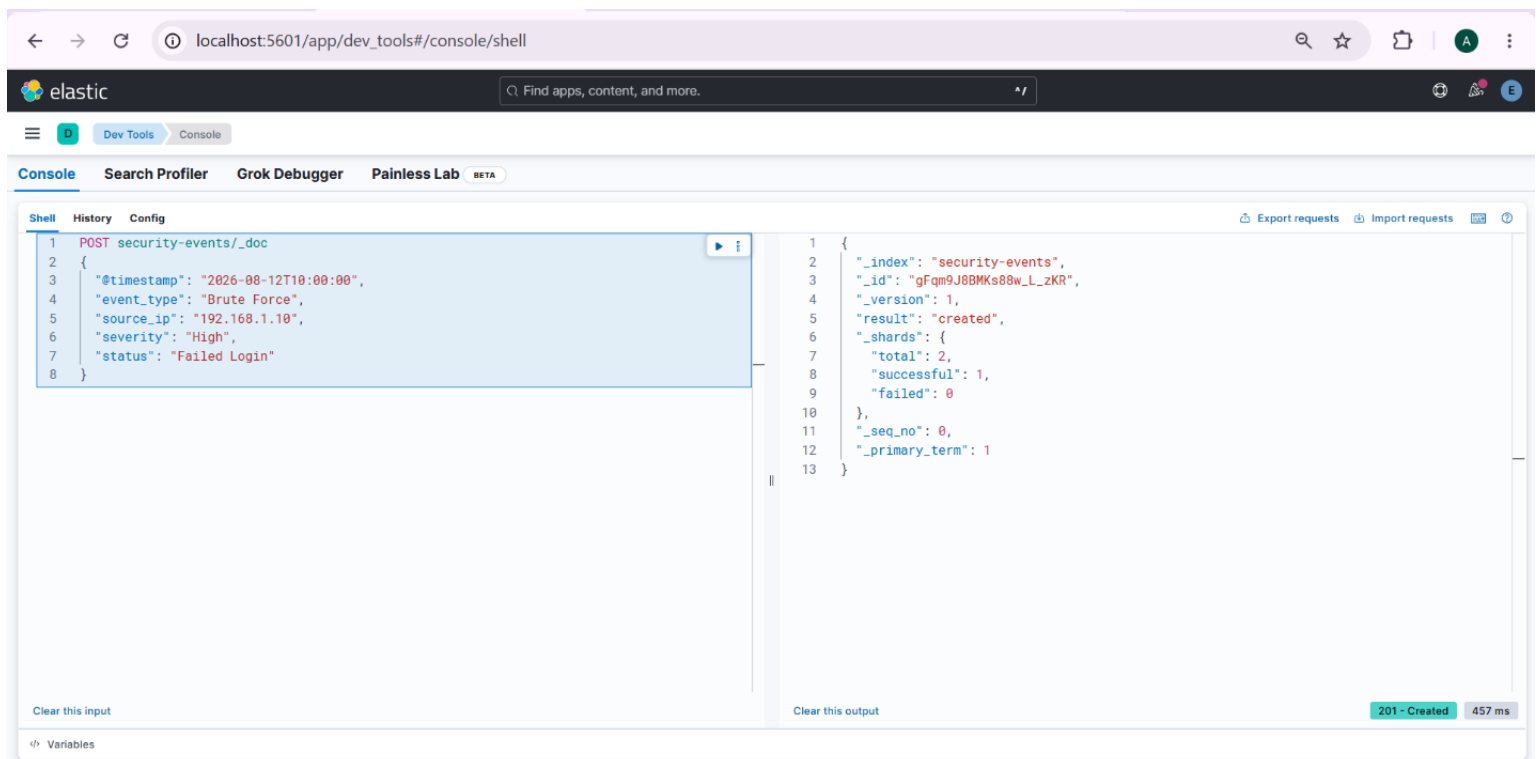


Figure 5.1: Insertion of simulated brute-force attack events into the Elasticsearch security-events index.

Observation:

Security events representing brute-force attack attempts were manually inserted into the security-events index using Elasticsearch Dev Tools. The records included timestamps, source IP addresses, severity levels, and failed authentication activities, providing the necessary dataset for detection rule testing.

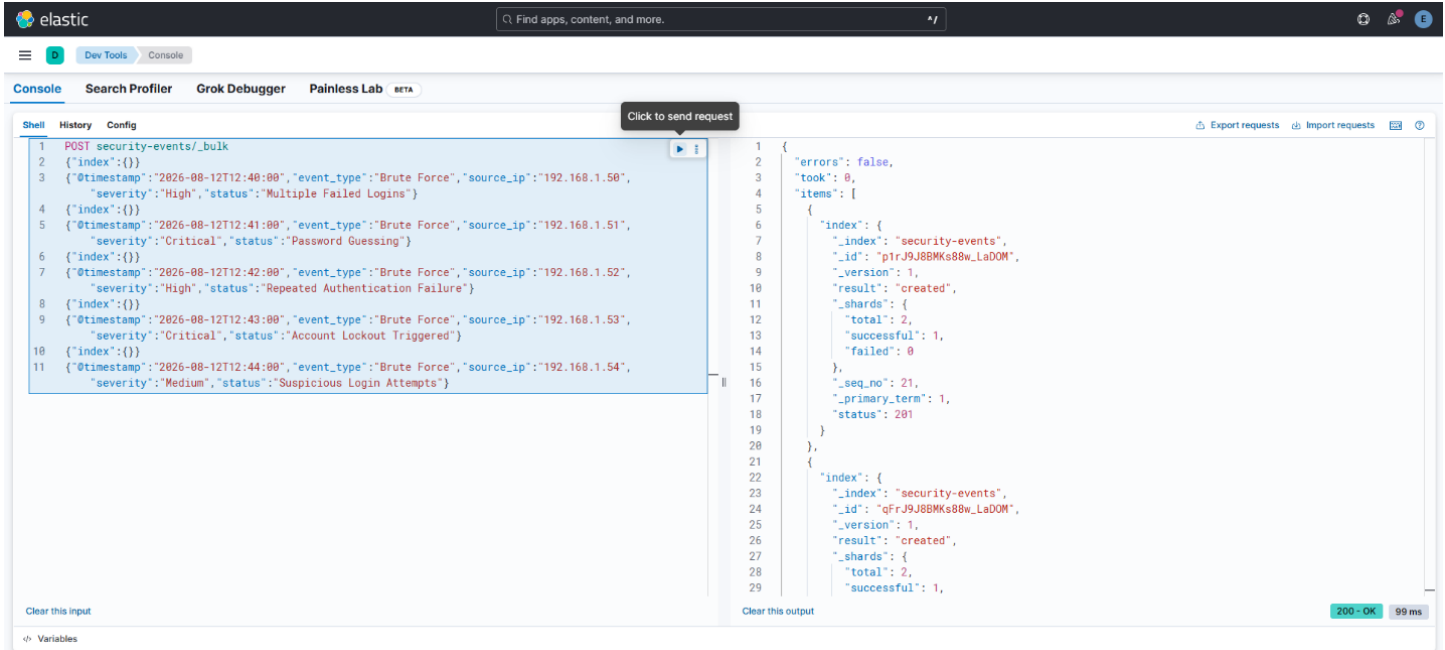


Figure 5.2: Bulk generation of brute-force attack events for security monitoring and detection testing.

Observation:

Multiple brute-force attack records with varying severity levels and authentication failure conditions were ingested into Elasticsearch using the Bulk API. This enabled the creation of a realistic attack dataset for validating detection rule functionality and alert generation.

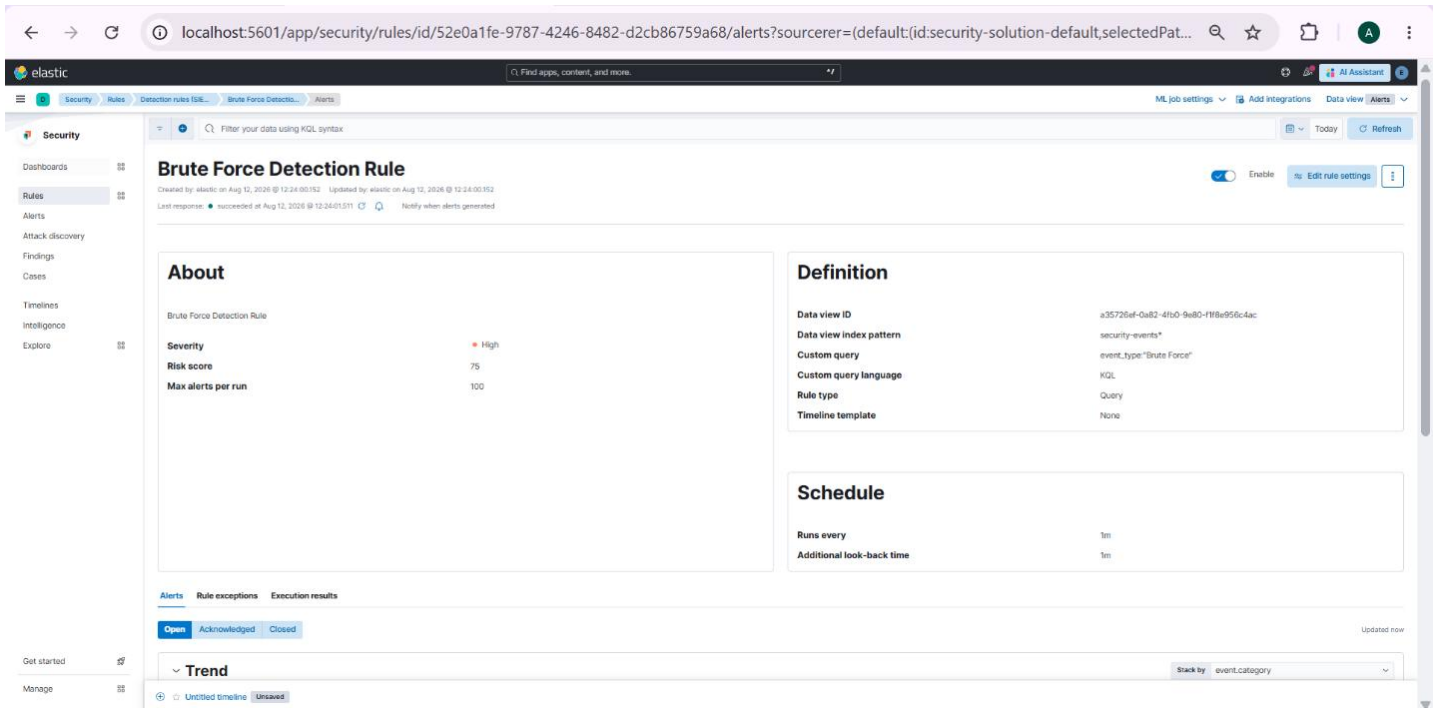


Figure 5.3: Configuration and execution of the Brute Force Detection Rule in Elastic Security.

Observation:

A custom query-based detection rule was created to identify events categorized as "Brute Force" within the security-events index. The rule was configured with a high severity level and scheduled execution interval to continuously monitor authentication-related attack activities.

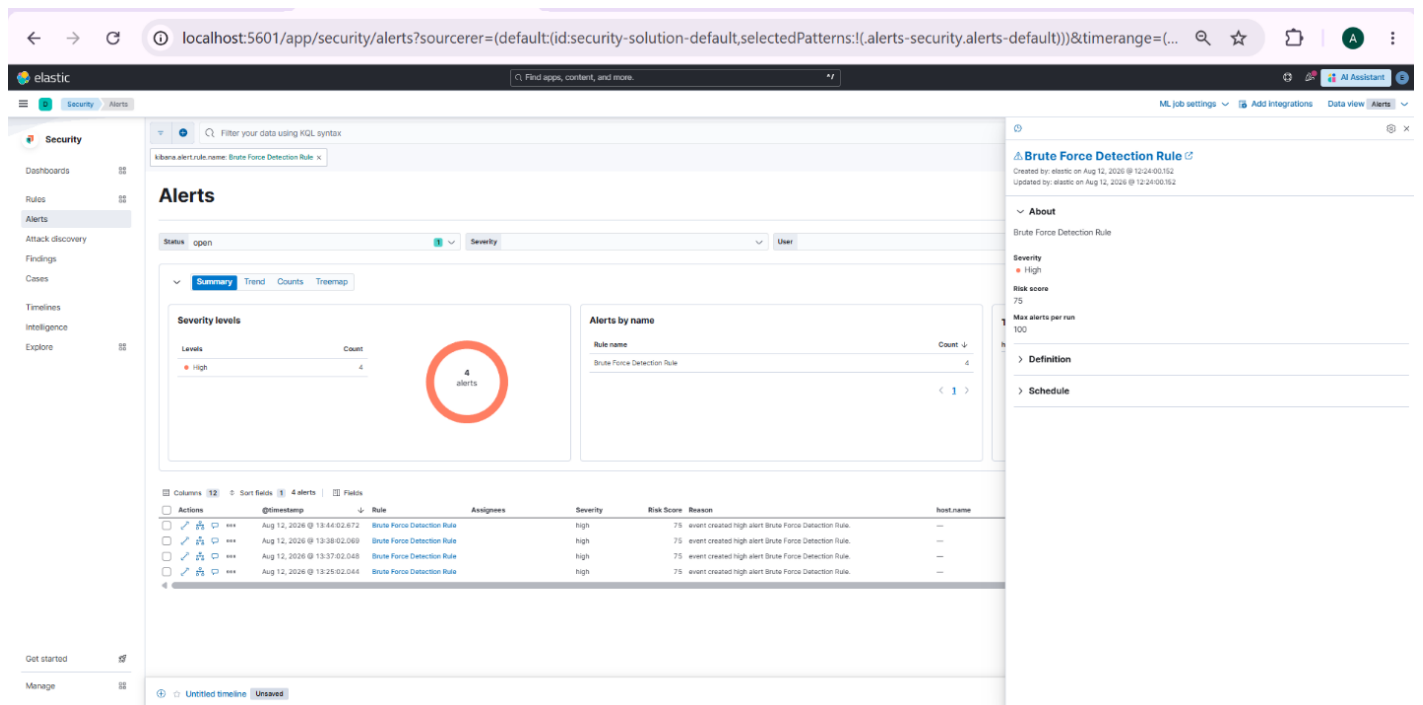


Figure 5.4: Security alerts generated following successful execution of the Brute Force Detection Rule.

Observation:

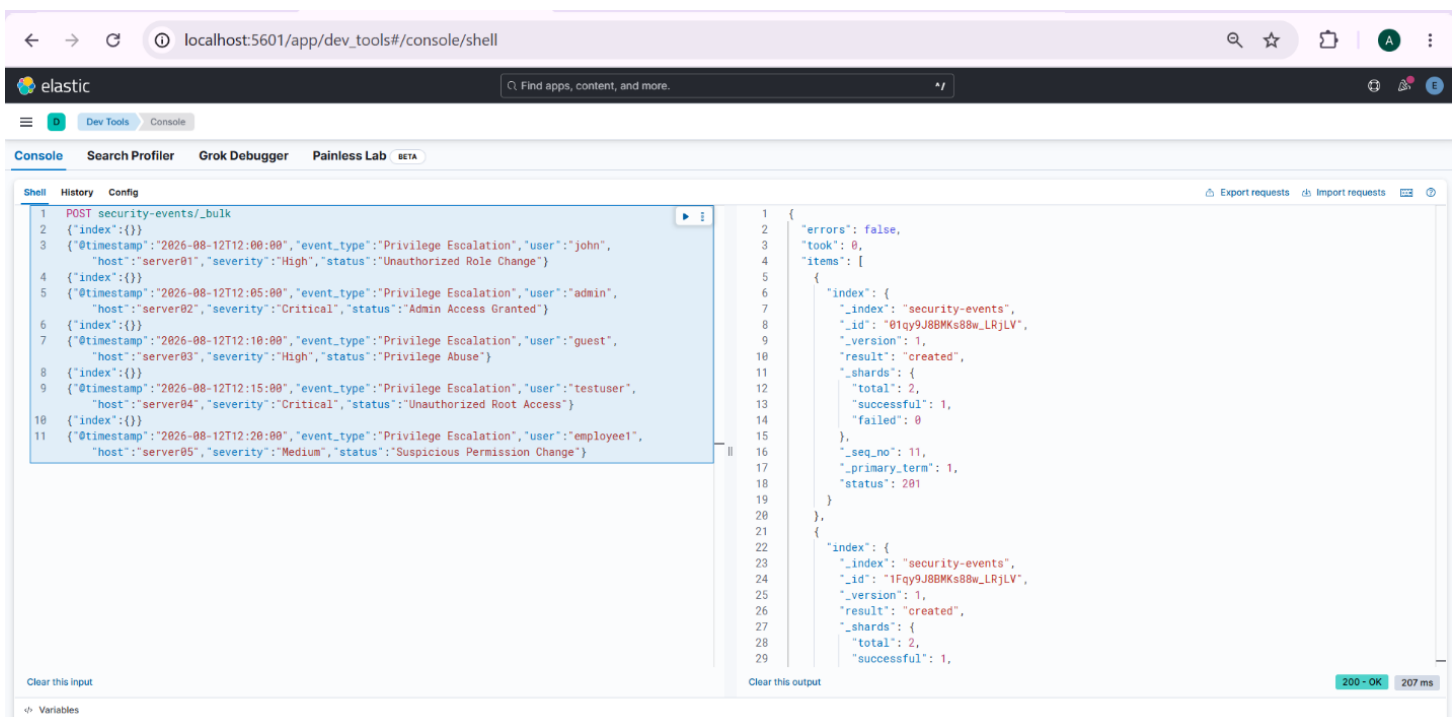
The configured detection rule successfully identified brute-force attack events from the ingested dataset and generated corresponding security alerts. The alert dashboard provided visibility into detected incidents, confirming the effectiveness of the implemented detection mechanism.

5. Chapter Summary

Simulated brute-force attack events were successfully ingested into Elasticsearch and monitored using a custom Elastic Security detection rule. The rule accurately identified authentication attack patterns and generated security alerts, demonstrating the capability of the SIEM platform to detect unauthorized access attempts and support proactive security monitoring.

6. Privilege Escalation Detection

Privilege escalation is a security attack technique in which an attacker gains higher-level permissions than originally assigned. Successful privilege escalation can allow unauthorized access to sensitive resources, administrative functions, and critical system configurations. Detecting such activities is essential for preventing system compromise and maintaining the integrity of organizational assets. In this implementation, simulated privilege escalation events were generated and monitored using Elastic Security to validate detection and alerting capabilities.



```
1 POST security-events/_bulk
2 {"index":{}}
3 {"@timestamp":"2026-08-12T12:00:00","event_type":"Privilege Escalation","user":"john",
4  "host":"server01","severity":"High","status":"Unauthorized Role Change"}
5 {"index":{}}
6 {"@timestamp":"2026-08-12T12:05:00","event_type":"Privilege Escalation","user":"admin",
7  "host":"server02","severity":"Critical","status":"Admin Access Granted"}
8 {"index":{}}
9 {"@timestamp":"2026-08-12T12:10:00","event_type":"Privilege Escalation","user":"guest",
10 "host":"server03","severity":"High","status":"Privilege Abuse"}
11 {"index":{}}
12 {"@timestamp":"2026-08-12T12:15:00","event_type":"Privilege Escalation","user":"testuser",
13 "host":"server04","severity":"Critical","status":"Unauthorized Root Access"}
14 {"index":{}}
15 {"@timestamp":"2026-08-12T12:20:00","event_type":"Privilege Escalation","user":"employee1",
16 "host":"server05","severity":"Medium","status":"Suspicious Permission Change"}

1 {
2   "errors": false,
3   "took": 0,
4   "items": [
5     {
6       "index": {
7         "_index": "security-events",
8         "_id": "01qy9J8BMks88w_LRjLV",
9         "_version": 1,
10        "result": "created",
11        "_shards": {
12          "total": 2,
13          "successful": 1,
14          "failed": 0
15        }
16      },
17      "_seq_no": 11,
18      "_primary_term": 1,
19      "status": 201
20    },
21    {
22      "index": {
23        "_index": "security-events",
24        "_id": "1Fqy9J8BMks88w_LRjLV",
25        "_version": 1,
26        "result": "created",
27        "_shards": {
28          "total": 2,
29          "successful": 1,
```

Figure 6.1: Insertion of simulated privilege escalation events into the Elasticsearch security-events index.

Observation:

Security events representing privilege escalation activities were successfully ingested into Elasticsearch. The event records included timestamps, source IP addresses, user information, severity classifications, and event types required for security monitoring and analysis.

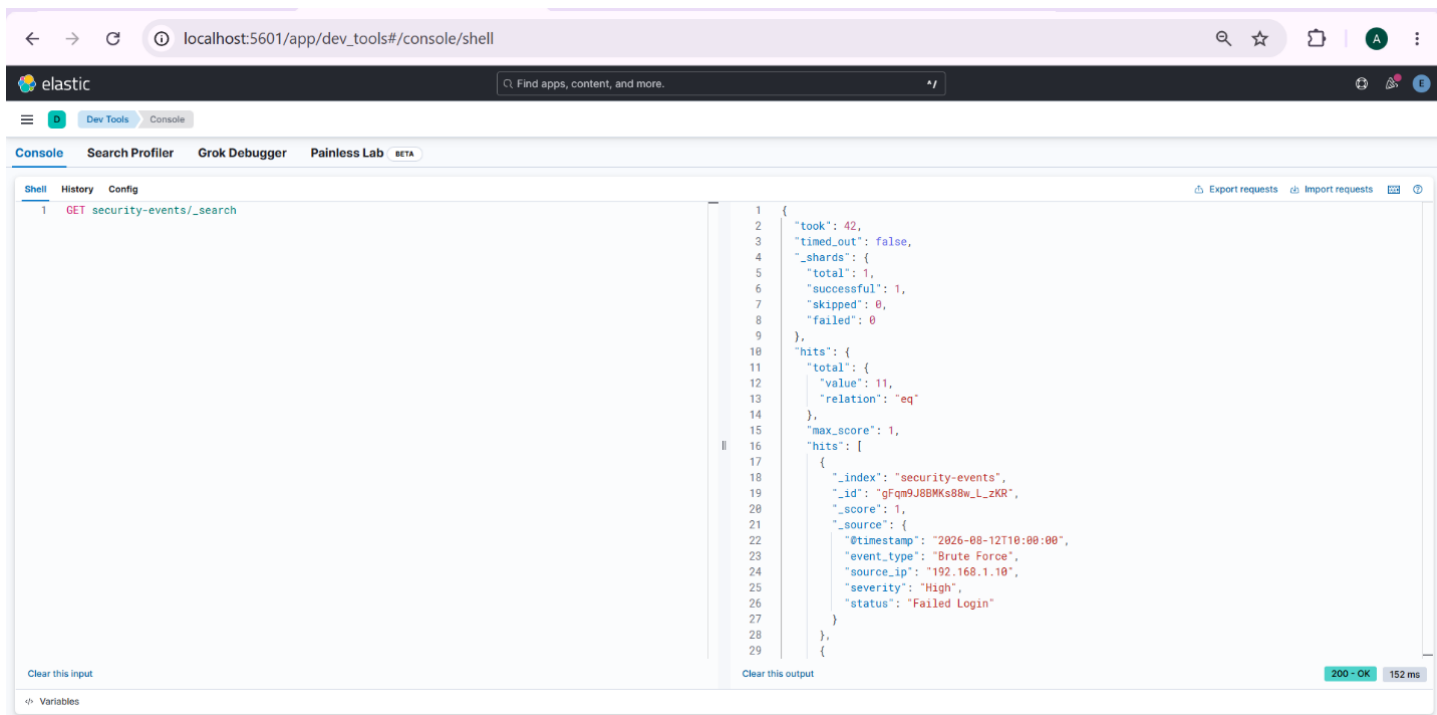


Figure 6.2: Privilege Escalation Events Verification

Observation:

The search results confirmed that the privilege escalation events were successfully stored and were retrievable from the security-events index. The indexed records provided the necessary data required for detection rule creation, alert generation, and dashboard visualization.

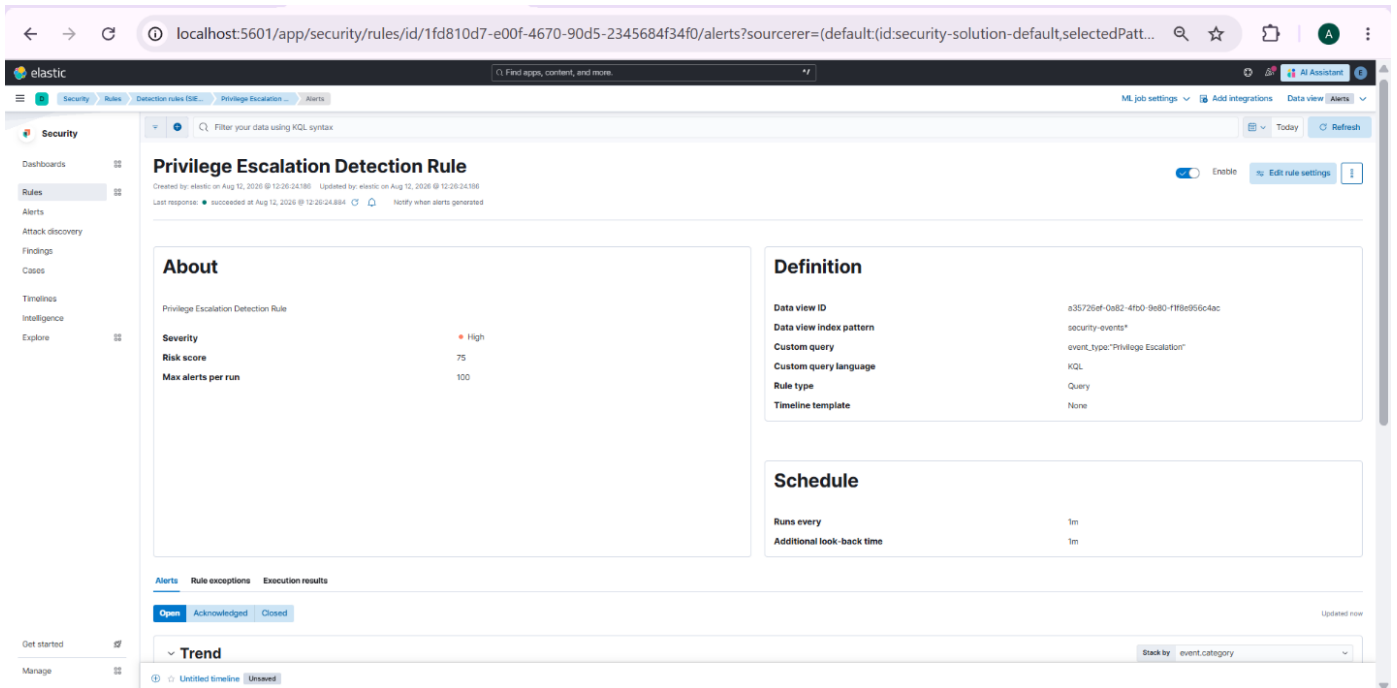


Figure 6.3: Privilege Escalation Detection Rule Configuration

Observation:

The detection rule was successfully configured and enabled within Elastic Security. The rule continuously monitored ingested privilege escalation events and provided the foundation for automated alert generation whenever matching activities were detected.

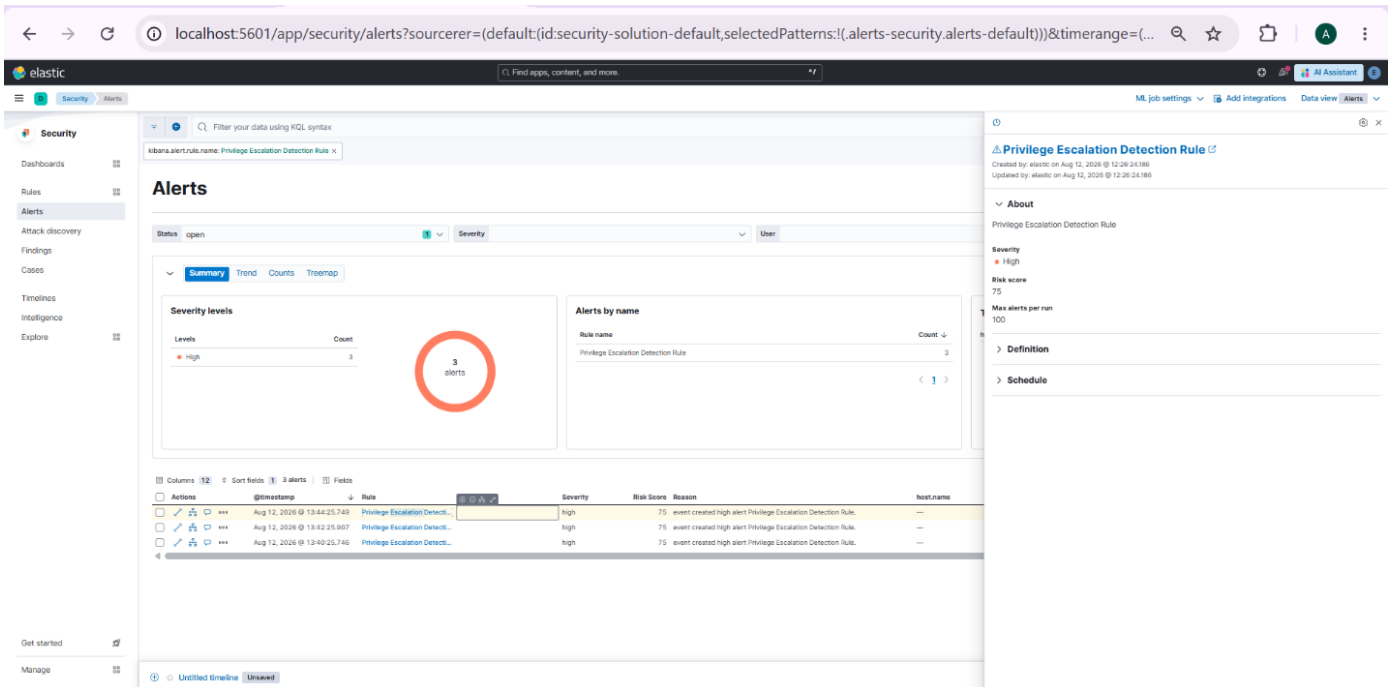


Figure 6.4: Privilege Escalation Alert Generation

Observation:

The detection rule successfully identified simulated privilege escalation activities and generated corresponding security alerts. The alert generation process validated the effectiveness of the rule configuration and demonstrated the capability of the SIEM platform to detect and monitor privilege abuse attempts in real time.

6. Chapter Summary

Privilege escalation events were successfully generated, ingested, verified, and monitored within the Elastic Security environment. A custom detection rule was configured to identify unauthorized privilege-related activities, and the rule successfully generated alerts for matching events. The successful detection and alert generation process demonstrated the effectiveness of the ELK Stack in monitoring, identifying, and analyzing privilege escalation attempts within a SIEM environment.

7. Data Exfiltration Attempts

Data exfiltration is a cyberattack technique in which sensitive information is transferred from an organization's environment to an unauthorized external destination. Attackers often use data exfiltration to steal confidential business information, intellectual property, credentials, or customer records. Detecting unusual data transfer activities is essential for preventing data breaches and minimizing organizational risk. In this implementation, simulated data exfiltration events were generated and monitored using Elastic Security to validate detection and alerting capabilities.

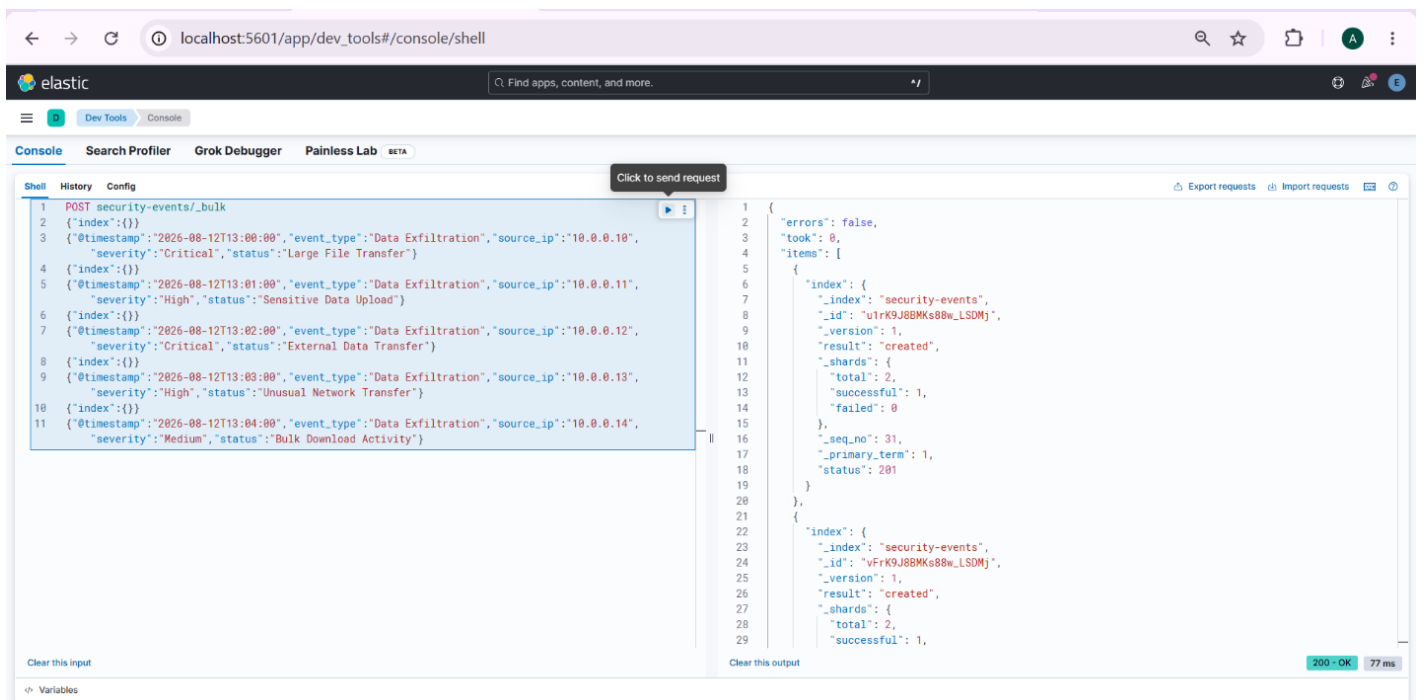


Figure 7.1: Data Exfiltration Event Ingestion

Observation:

The bulk ingestion process completed successfully, and all data exfiltration events were indexed without errors. The generated records became available for rule-based detection, security analysis, and dashboard visualization.

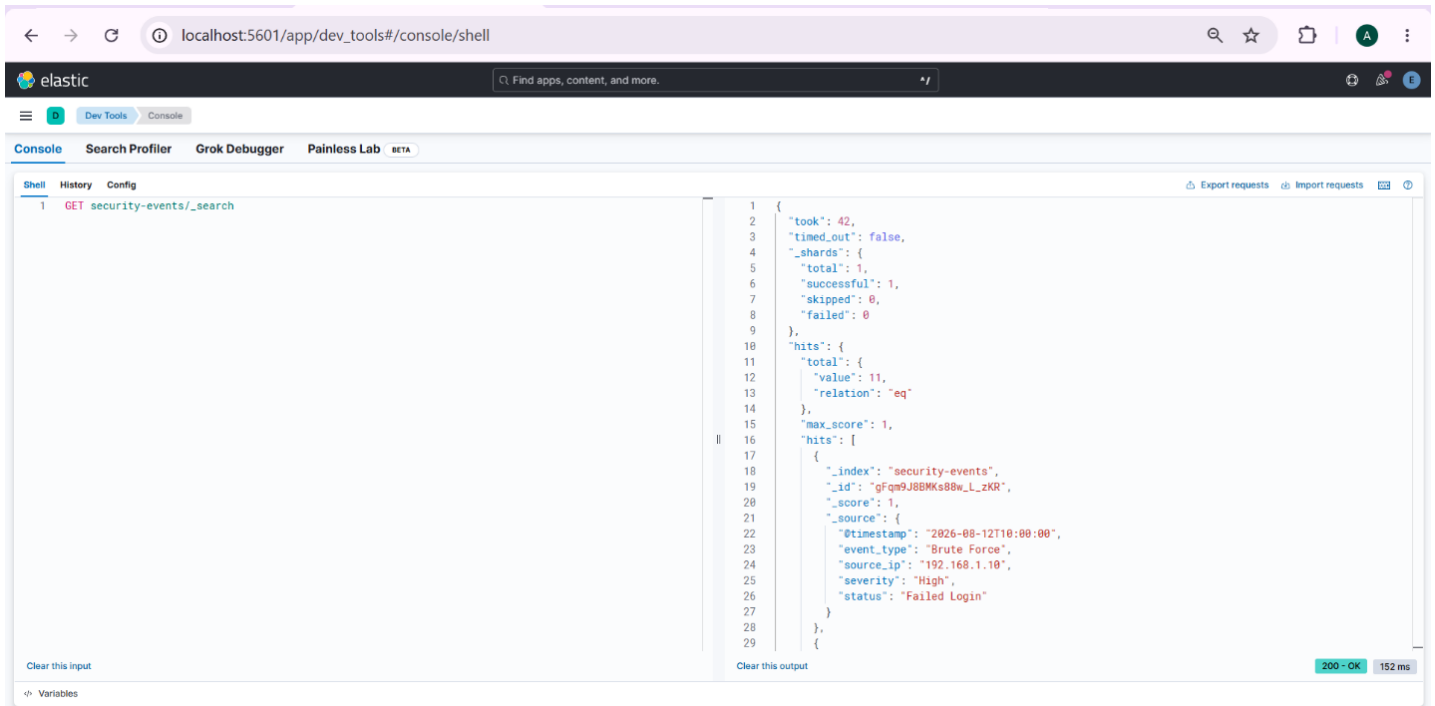


Figure 7.2: Data Exfiltration Events Verification

Observation:

The search results confirmed successful indexing of the data exfiltration events. All generated records were accessible for monitoring, rule execution, alert generation, and security investigation activities.

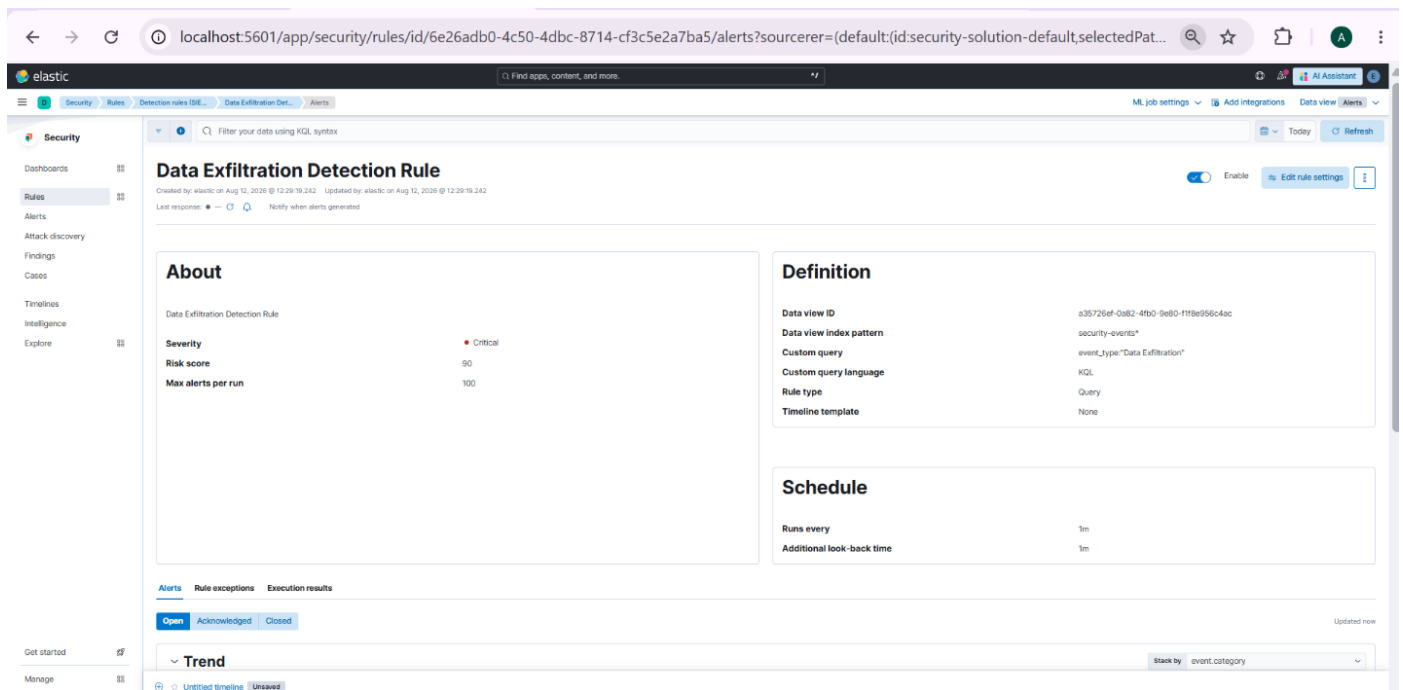


Figure 7.3: Data Exfiltration Detection Rule Configuration

Observation:

The detection rule was successfully created and enabled. The configuration ensured continuous monitoring of exfiltration-related activities and provided automated detection capabilities for suspicious data transfer events.

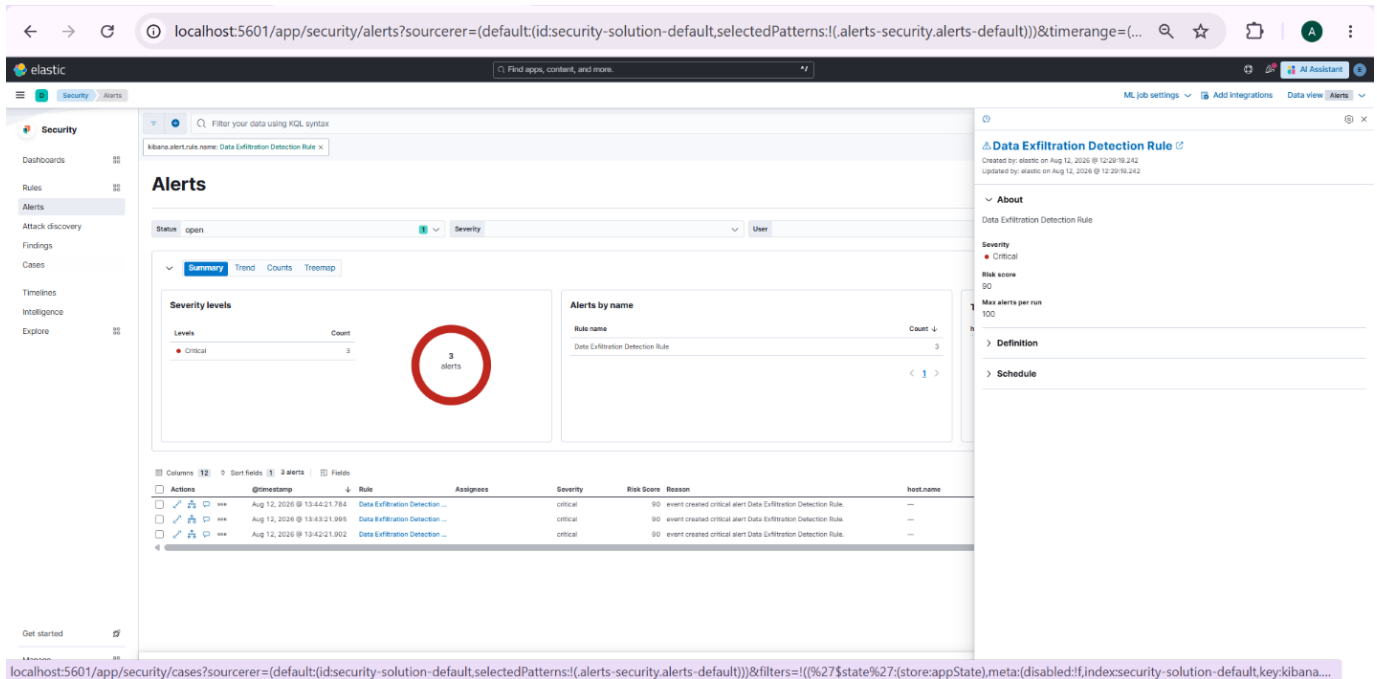


Figure 7.4: Data Exfiltration Alert Generation

Observation:

The configured detection rule successfully identified simulated data exfiltration activities and generated corresponding security alerts. This validated the effectiveness of the implemented detection logic and demonstrated the SIEM platform's capability to monitor potential data theft attempts.

7. Chapter Summary

Data exfiltration events were successfully generated, indexed, verified, and monitored within the Elastic Security environment. A custom detection rule was implemented to identify suspicious outbound data transfer activities, and the rule successfully generated alerts for matching events. The successful detection process demonstrated the effectiveness of the ELK Stack in monitoring and analyzing potential data exfiltration attempts within a SIEM environment.

8. Security Events Dashboard Development

After successfully generating and detecting security events related to brute-force attacks, privilege escalation activities, and data exfiltration attempts, a custom SIEM dashboard was developed using Kibana. The dashboard was designed to provide centralized visibility into security events, alert severity levels, attack categories, source IP activity, and recent security incidents. Multiple visualizations were created to support efficient security monitoring, event analysis, and incident investigation within a Security Operations Center (SOC) environment.

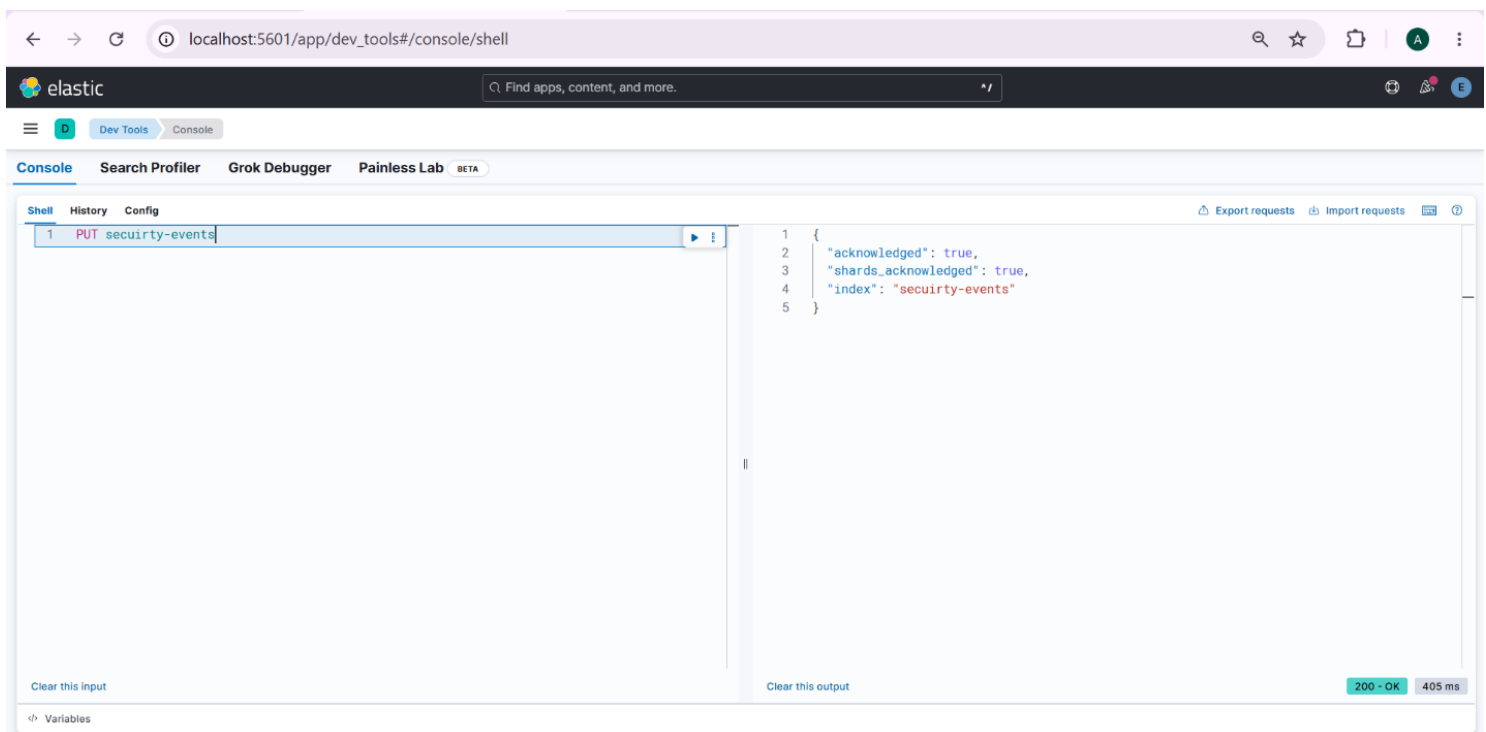


Figure 8.1: Creation of Security Events Index

Observation:

A dedicated Elasticsearch index named security-events was successfully created to serve as the primary repository for all simulated security event data. The index was designed to store records related to brute-force attacks, privilege escalation activities, and data exfiltration attempts. Establishing a dedicated index ensured structured data management, efficient search capabilities, and seamless integration with Kibana visualizations for security monitoring and analysis.

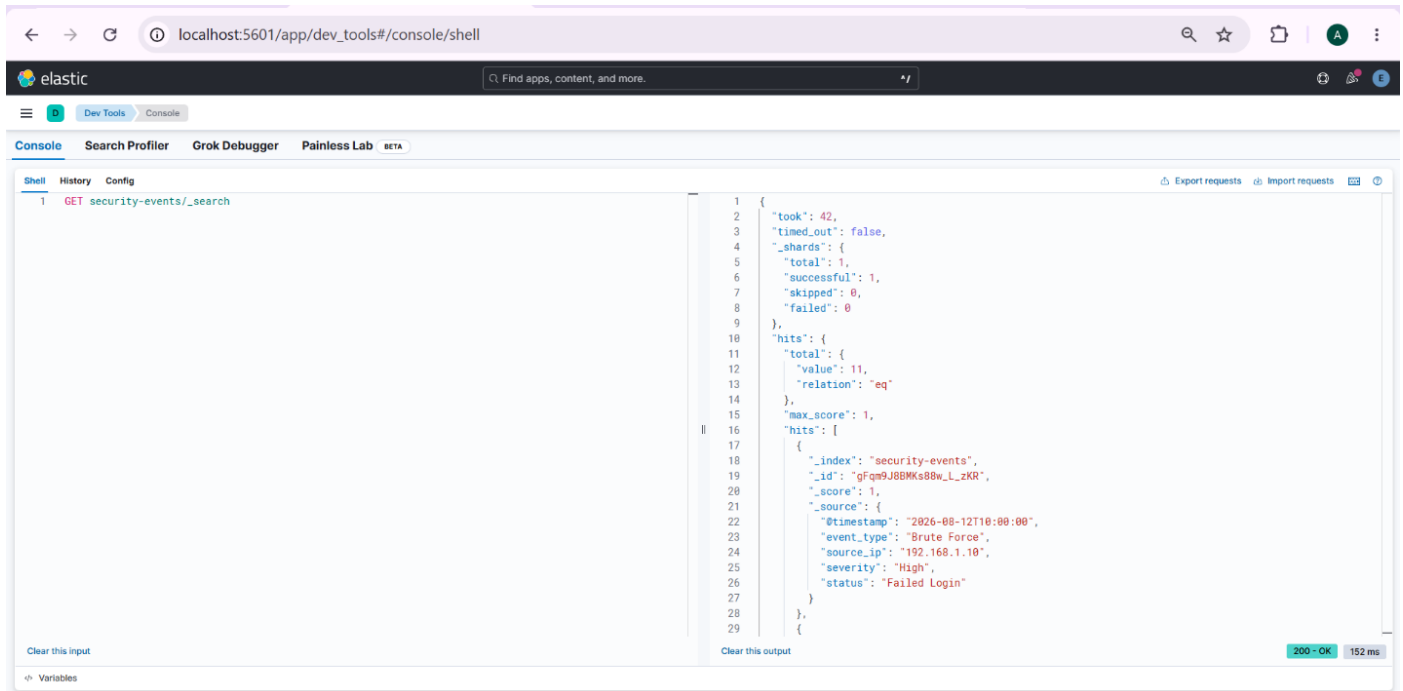


Figure 8.2: Security Events Data Verification

Observation:

The security event dataset was successfully verified using Elasticsearch search queries. The retrieved records confirmed that all generated events were correctly indexed and contained the required fields, including timestamps, event types, severity levels, source IP addresses, and user information. This validation step ensured data integrity and confirmed that the dataset was ready for dashboard development, visualization creation, and security event analysis.

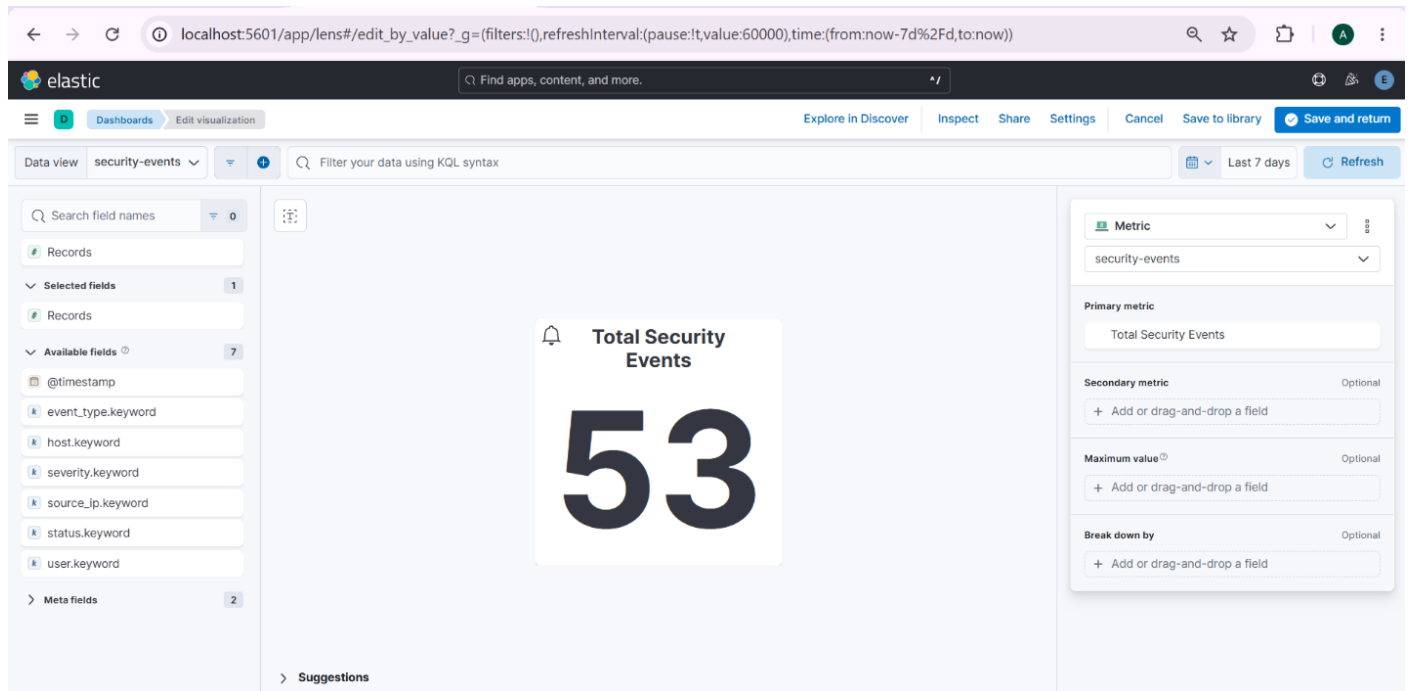


Figure 8.3: Total Security Events Metric Visualization

Observation:

The Total Security Events metric successfully displayed the aggregated count of all indexed security events. This visualization provides a high-level summary of the security dataset and enables analysts to quickly assess the overall volume of detected activities. Such metrics are commonly used in SOC environments to monitor event growth and evaluate the current security posture of monitored systems.

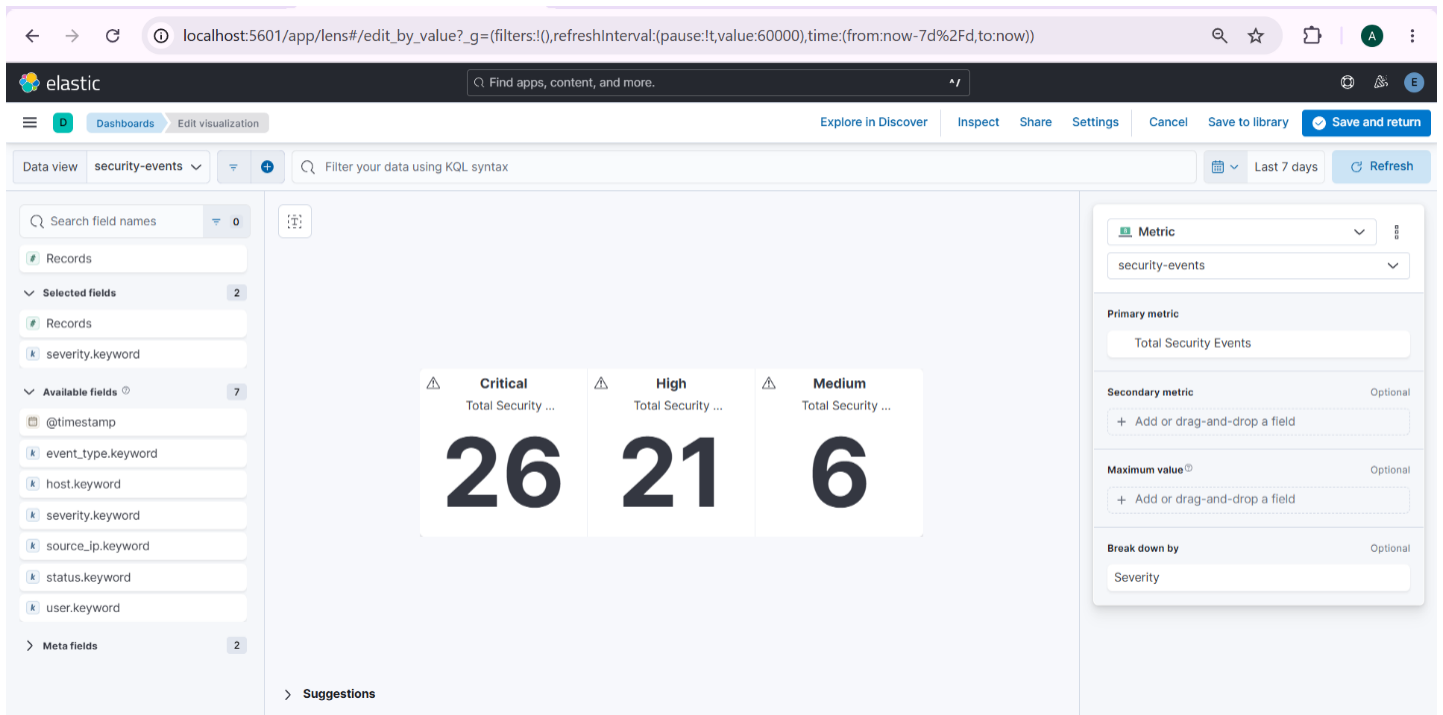


Figure 8.4: Severity-Based Security Metrics

Observation:

Severity-based metric cards provided a clear breakdown of security events according to their assigned risk levels. The visualization enabled rapid identification of high-priority incidents requiring immediate attention while also maintaining visibility into medium-risk activities. This approach improves incident prioritization and supports efficient security operations by highlighting the most significant threats within the environment.

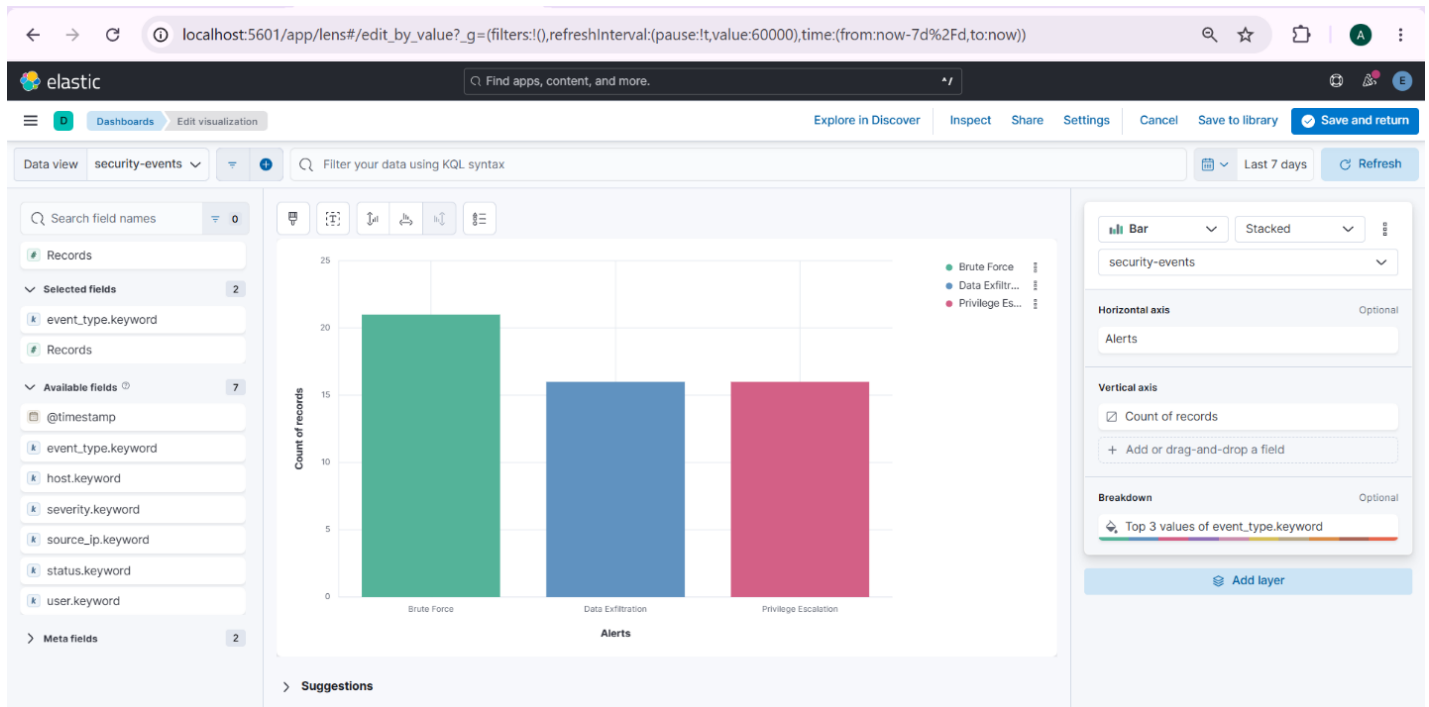


Figure 8.5: Attack Type Distribution Visualization

Observation:

The Attack Type Distribution visualization provided a clear comparison of the different security incidents present within the environment. By categorizing events according to attack type, analysts can quickly identify the most prevalent threats and understand the overall attack landscape. This visualization improves situational awareness and supports security teams in prioritizing monitoring and response efforts based on observed attack patterns.

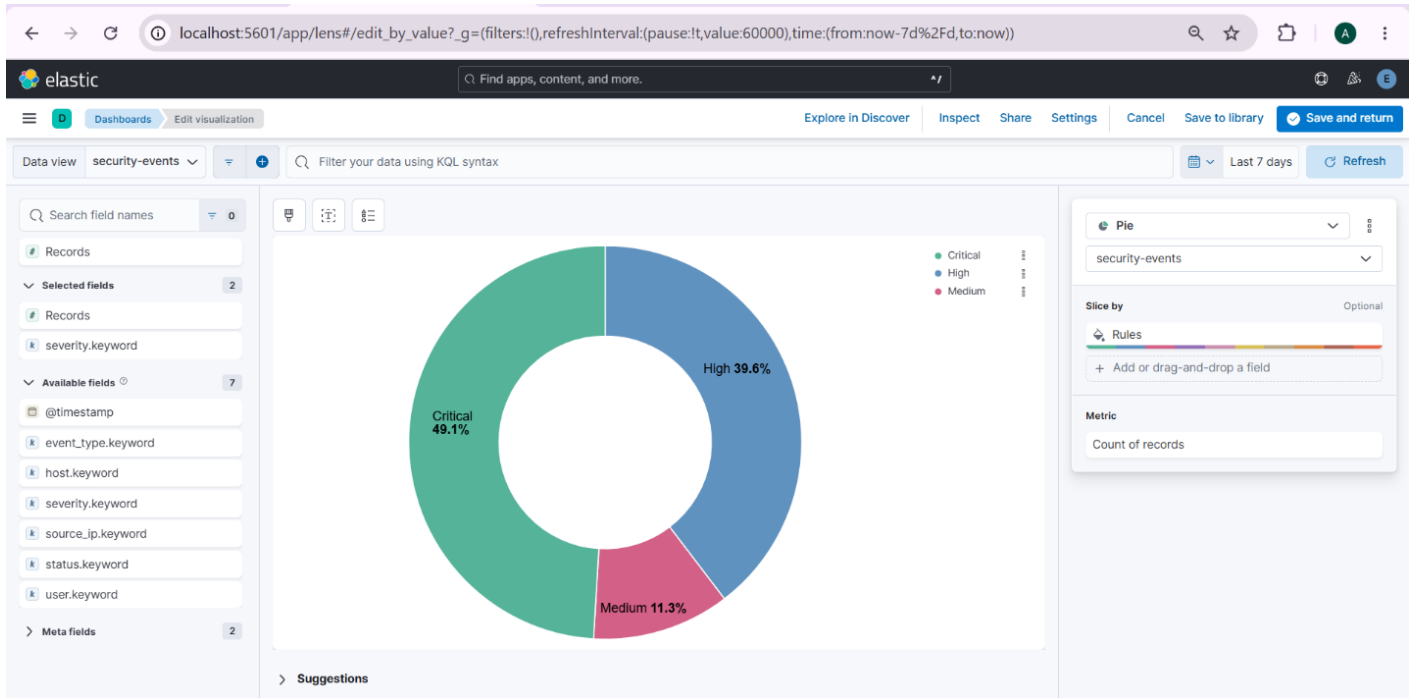


Figure 8.6: Alert Severity Distribution Visualization

Observation:

The severity distribution chart provided an effective overview of the risk profile of the collected security events. The visualization highlighted the proportion of Critical, High, and Medium severity incidents, enabling analysts to quickly assess threat levels and prioritize investigation efforts. Such severity-based visualizations are widely used in SOC environments to support risk-based decision-making and incident response activities.

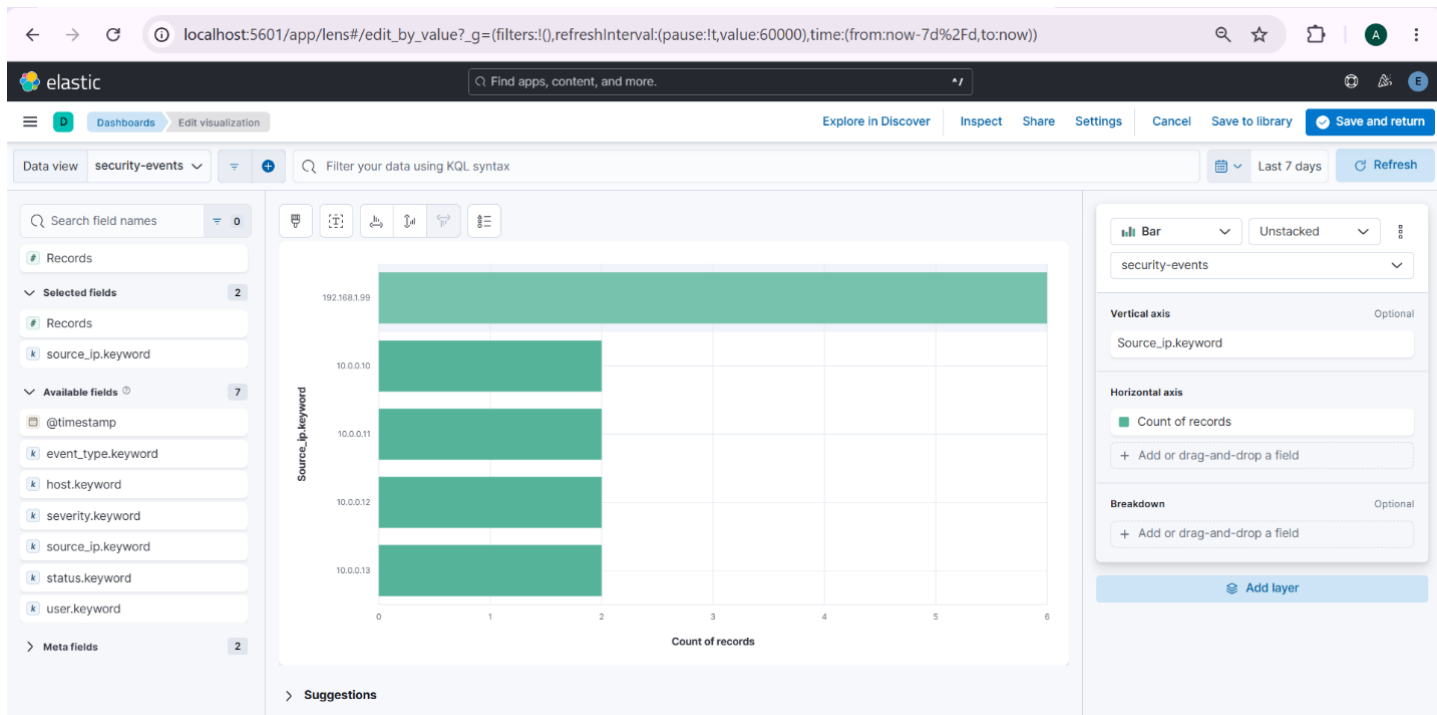


Figure 8.7: Top Source IPs by Alert Count

Observation:

The Top Source IPs visualization provided valuable insight into the origin of security events within the dataset. By highlighting IP addresses associated with the highest alert volumes, the dashboard enables analysts to focus investigations on systems exhibiting unusual or potentially malicious behavior. This analysis supports threat hunting and assists in identifying recurring sources of security incidents.

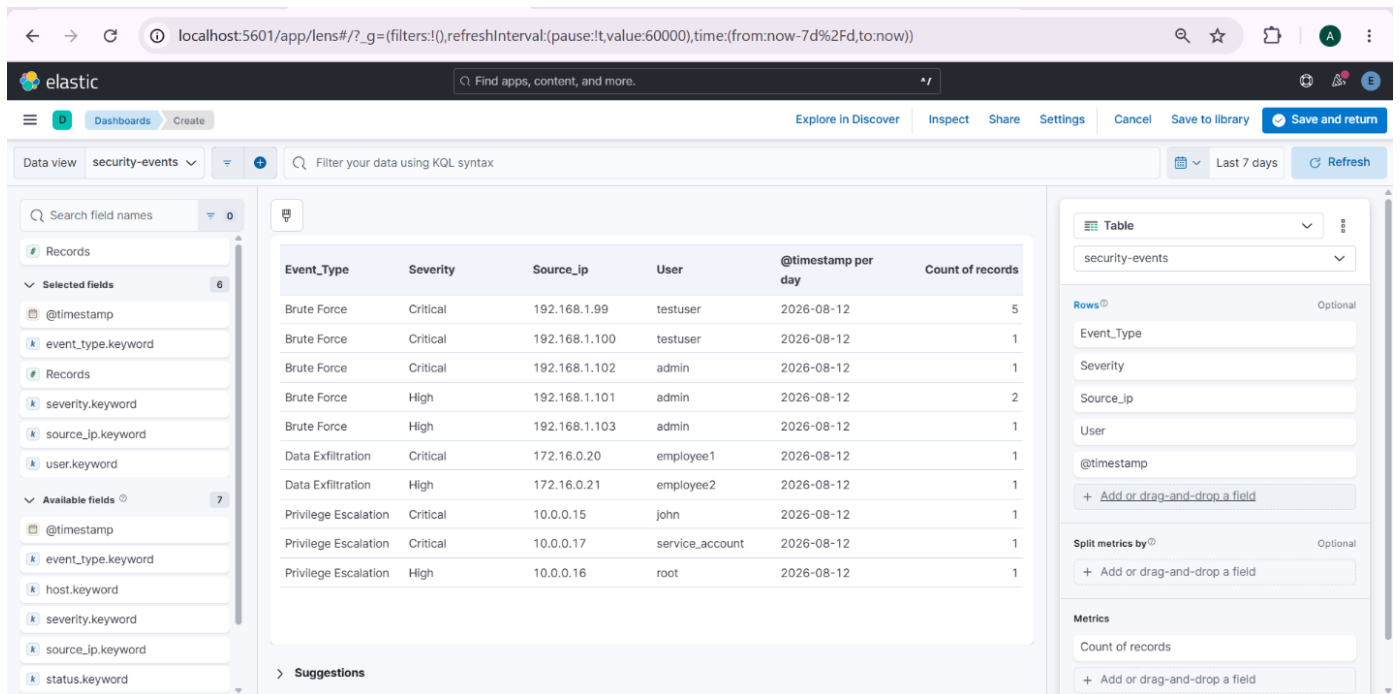


Figure 8.8: Recent Security Events Monitoring Table

Observation:

The Recent Security Events table provided detailed visibility into individual security records and supported efficient event investigation. Unlike aggregated charts, the tabular view allowed analysts to examine specific event attributes and correlate security activities across multiple indicators. This visualization serves as an important component of operational security monitoring and incident response workflows.

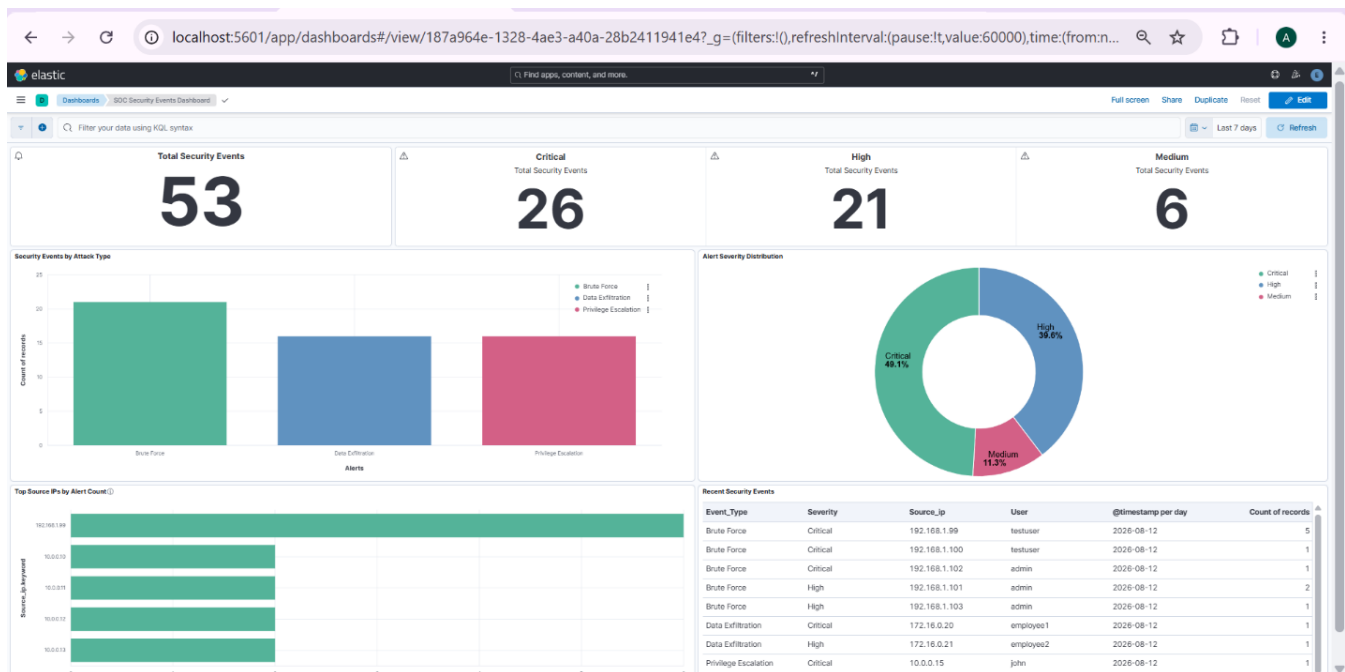


Figure 8.9: Final SOC Security Events Dashboard

Observation:

The final dashboard successfully consolidated critical security information into a single monitoring platform. By combining high-level metrics with detailed analytical visualizations, the dashboard improved visibility into security events, facilitated threat identification, and supported efficient incident investigation. The implementation demonstrates how the ELK Stack can be utilized as a SIEM platform for security monitoring, event analysis, and operational decision-making within a SOC environment.

8.1 Chapter Summary

The Security Events Dashboard was successfully developed using Kibana and integrated with Elasticsearch to provide centralized visibility into security activities. Multiple visualizations, including metric cards, attack distribution charts, severity analysis, source IP monitoring, and event tracking tables, were created to support security monitoring and investigation. The completed dashboard transformed raw security event data into meaningful insights, enabling efficient analysis, threat detection, and situational awareness within the monitored environment.

9. Testing And Validation

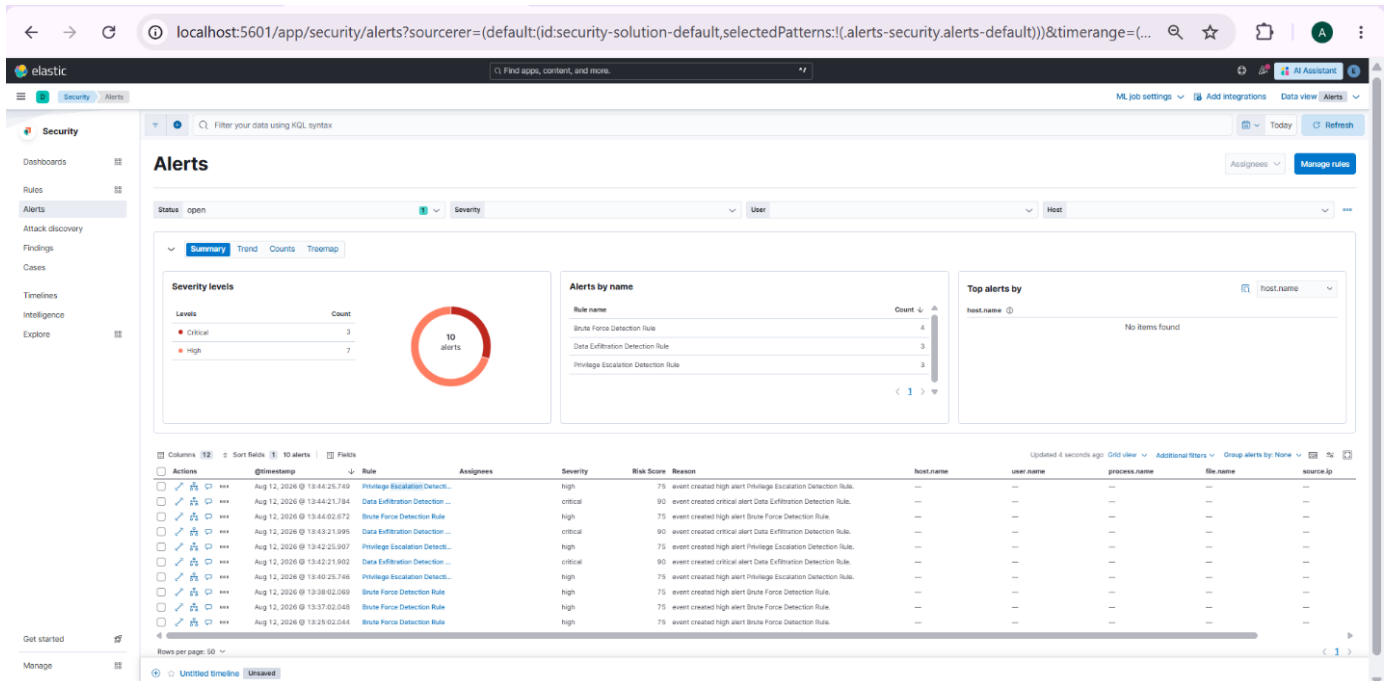


Figure 9.1: Security Alerts Validation

Observation:

The alerts dashboard confirmed that all detection rules executed successfully and generated the expected security alerts. A total of ten alerts were recorded, including both High and Critical severity incidents. This validation demonstrated the effectiveness of the implemented detection logic and verified the successful operation of the alerting mechanism within the SIEM environment.

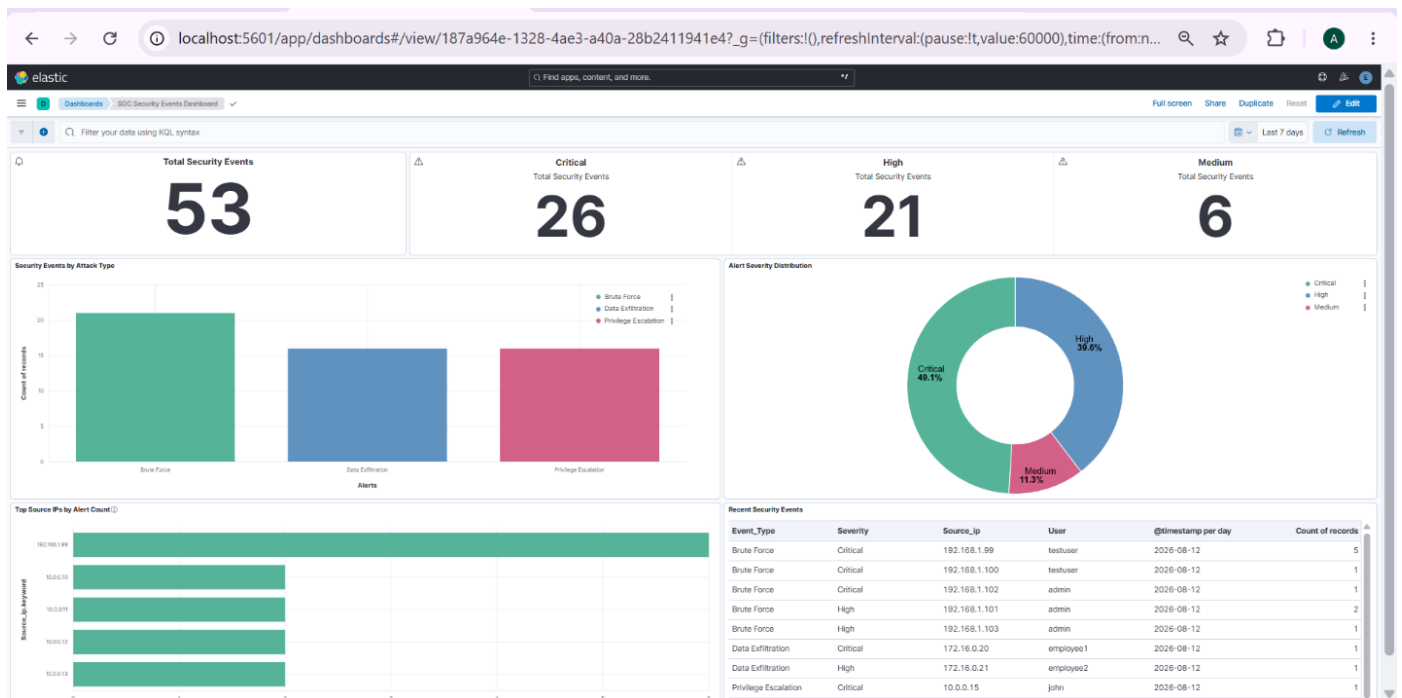


Figure 9.2: Final Security Events Dashboard Validation

Observation:

The dashboard successfully visualized all generated security events and accurately reflected the results of the implemented detection rules. Metric cards, charts, and event tables displayed consistent and reliable data, confirming successful integration between Elasticsearch, Kibana, and Elastic Security. The validation process demonstrated the dashboard's capability to provide real-time visibility into security activities and support effective security monitoring.

9.1 Chapter Summary

Testing and validation confirmed the successful ingestion, detection, alert generation, and visualization of simulated security events. The implemented detection rules accurately identified brute-force attacks, privilege escalation activities, and data exfiltration attempts, while the alerts dashboard and Kibana visualizations provided comprehensive visibility into security incidents. The successful operation of all components demonstrated the effectiveness of the ELK Stack as a SIEM solution for security monitoring, event analysis, and incident detection.

10. Conclusion

The objective of designing and implementing a custom SIEM dashboard using the ELK Stack was successfully achieved. Security events representing brute-force attacks, privilege escalation activities, and data exfiltration attempts were generated, indexed, monitored, and analyzed within the Elastic Security environment. Custom detection rules were successfully configured to identify security incidents and generate corresponding alerts for each attack category. Furthermore, a centralized Kibana dashboard was developed to visualize critical security information through metric cards, attack distribution charts, severity-based analytics, source IP monitoring, and event tracking tables. The dashboard transformed raw security event data into meaningful insights, enabling efficient monitoring, threat detection, and incident investigation.

Overall, this project provided practical experience in SIEM implementation, security event analysis, detection engineering, and dashboard development using the ELK Stack. The completed solution demonstrates the effectiveness of Elasticsearch and Kibana in supporting Security Operations Center (SOC) activities and improving organizational security visibility.